

First-Cut Prototype

2-Day Build Plan (Phases 1–4 of the Full Roadmap, Compressed)

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Scope & Approach

This is a vertical slice, not the full system: it compresses Phase 1 (Receipt Extraction), Phase 2 (Policy Validation), a light version of Phase 3 (Anomaly Detection), and a simulated version of Phase 4 (Workflow Routing) from the full roadmap into roughly 14.5 focused hours across 2 days. It proves the core Claude-powered extraction-plus-reasoning loop end to end.

Out of scope for this build: real ERP/HRIS integration (Phase 5), a production employee portal (Phase 6), and pilot hardening (Phase 7) — those depend on real enterprise systems and access that a 2-day prototype doesn't have. The prototype uses purpose-built dummy datasets in place of live systems; each phase below states exactly what dummy data it needs and, where new data is introduced, its contents.

2-Day Timeline

Phase	Title	Day	Time	Duration
1	Environment & Dummy Data Foundation	Day 1	Hour 0.0–1.5	1.5 hrs
2	Receipt Capture & Claude Extraction	Day 1	Hour 1.5–4.5	3 hrs
3	Policy Validation Engine	Day 1	Hour 4.5–7.0	2.5 hrs
4	Anomaly & Duplicate Flagging	Day 2	Hour 0.0–2.0	2 hrs
5	Workflow Decision Simulation	Day 2	Hour 2.0–4.0	2 hrs
6	Demo Interface	Day 2	Hour 4.0–6.5	2.5 hrs
7	End-to-End Test & Packaging	Day 2	Hour 6.5–7.5	1 hr

Total: ~14.5 focused hours across 2 days (~7.25 hrs/day), leaving buffer for setup and troubleshooting.

Phase-by-Phase Plan

Phase 1 — Environment & Dummy Data Foundation (Day 1, Hour 0.0–1.5)

Objective	Stand up the project skeleton and produce every reference dataset the rest of the prototype depends on, since there is no live ERP/HRIS access for a first-cut build.
Build Tasks	- Create project scaffold: /data, /scripts, /output folders - Build employee_roster.csv — 10 employees across Sales, Engineering, Marketing, Executive, with reporting managers and cost centers - Build policy_rulebook.json — 10 structured rules covering meals, alcohol, lodging, airfare, ground transport, mileage, per-diem, office supplies, receipt threshold, and submission window - Build approval_matrix.json — auto-approve / manager / finance routing thresholds
Dummy Dataset	employee_roster.csv (10 rows)

	● policy_rulebook.json (10 rules) ● approval_matrix.json (4 routing conditions)
Claude's Role	None yet — this phase is pure data setup.
Deliverable / Exit Criteria	Three reference datasets created and spot-checked for realism.
Duration	1.5 hrs

Phase 2 — Receipt Capture & Claude Extraction *(Day 1, Hour 1.5–4.5)*

Objective	Prove the core "unstructured receipt → structured transaction" loop, the foundation everything downstream depends on.
Build Tasks	● Author 12 dummy raw receipts (raw_receipts.json) simulating photographed/emailed/scanned content — restaurant, taxi, hotel, airfare, office supplies, mileage log, a low-quality/ambiguous scan, and deliberately seeded edge cases (see dataset) ● Prompt Claude with structured JSON output to extract vendor, date, amount, currency, category, line items, and a confidence score from each raw receipt ● Persist results to transactions.json
Dummy Dataset	● raw_receipts.json (12 receipts) — seeded with: RCPT-001 (alcohol on a meal receipt, a policy test case), RCPT-006 & RCPT-011 (identical transaction submitted twice via different channels, a duplicate test case), RCPT-010 (low-quality scan with an ambiguous digit, a confidence-scoring test case), RCPT-012 (a solo lunch priced far above the meal cap, an anomaly test case)
Claude's Role	Multimodal/text extraction into structured JSON — maps to Phase 1 of the full roadmap.
Deliverable / Exit Criteria	transactions.json with 12 structured, confidence-scored records.
Duration	3 hrs

Phase 3 — Policy Validation Engine *(Day 1, Hour 4.5–7.0)*

Objective	Automatically adjudicate every transaction against the policy rulebook and attach a plain-language explanation to each decision.
Build Tasks	● Run a deterministic rule check (limits, receipt-required threshold, prohibited categories) against policy_rulebook.json ● Run a Claude reasoning pass for ambiguous cases a static rule can't resolve (e.g., is a \$145 taxi reasonable given the airport-to-hotel route and a traffic detour noted on the receipt) ● Attach a decision (compliant / flagged) and a human-readable reason to every record
Dummy Dataset	None new — reuses datasets from earlier phases

Claude's Role	Judgment-based policy reasoning and explanation generation — maps to Phase 2 of the full roadmap.
Deliverable / Exit Criteria	validated_transactions.json — every record carries a decision plus a reason. RCPT-001's alcohol line item should be correctly flagged.
Duration	2.5 hrs

Phase 4 — Anomaly & Duplicate Flagging *(Day 2, Hour 0.0–2.0)*

Objective	Demonstrate the "catch what static rules miss" layer using the duplicate and outlier cases seeded into the Day 1 dataset.
Build Tasks	● Hash/fuzzy-match duplicate detection on vendor + date + amount to catch RCPT-006/RCPT-011 ● Simple statistical outlier check — flag any amount more than 2x the category average across the 12-record set, which should catch RCPT-012 ● Claude pass to write a reasoned explanation for each flagged anomaly
Dummy Dataset	None new — reuses datasets from earlier phases
Claude's Role	Contextual anomaly reasoning — a lightweight version of Phase 3 of the full roadmap.
Deliverable / Exit Criteria	risk_scored_transactions.json — RCPT-006/011 flagged as duplicates, RCPT-012 flagged as an outlier.
Duration	2 hrs

Phase 5 — Workflow Decision Simulation *(Day 2, Hour 2.0–4.0)*

Objective	Simulate approval routing so the demo shows a complete decision end to end, not just an isolated flag.
Build Tasks	● Apply approval_matrix.json thresholds against each record's policy result and risk score to assign a status: auto-approved, pending manager approval, pending finance approval, or rejected ● Attach the correct approver from employee_roster.csv using each employee's reporting manager ● Have Claude draft a one-line approval summary per transaction for the simulated approver
Dummy Dataset	None new — reuses datasets from earlier phases
Claude's Role	Drafts a concise, plain-language approval summary per transaction — maps to Phase 4 of the full roadmap.
Deliverable / Exit Criteria	final_decisions.json — every one of the 12 transactions carries a routed status, an approver, and a summary.
Duration	2 hrs

Phase 6 — Demo Interface *(Day 2, Hour 4.0–6.5)*

Objective	Make the pipeline visible and demoable rather than leaving the output as raw JSON files.
Build Tasks	● Build a single-page dashboard: a transaction table (vendor, amount, category, decision, status, reason), filterable by status ● Click-through view on any row showing the record's path through extraction → policy check → risk flag → routing ● Load the precomputed final_decisions.json — no live API calls needed to run the demo itself
Dummy Dataset	None new — reuses datasets from earlier phases
Claude's Role	None directly — the dashboard visualizes the outputs Claude already produced in Phases 2–5.
Deliverable / Exit Criteria	A working demo dashboard showing all 12 transactions end to end.
Duration	2.5 hrs

Phase 7 — End-to-End Test & Packaging *(Day 2, Hour 6.5–7.5)*

Objective	Confirm the full pipeline runs cleanly top to bottom and package it for a walkthrough.
Build Tasks	● Re-run the full pipeline (data → extraction → validation → anomaly → routing → dashboard) from a clean state ● Spot-check that the seeded cases were caught correctly: alcohol violation, duplicate pair, ambiguous scan, inflated lunch ● Write a 1-page README / demo script — what to click, what to say, what each seeded case proves
Dummy Dataset	None new — reuses datasets from earlier phases
Claude's Role	None — verification and packaging.
Deliverable / Exit Criteria	A working, demoable prototype plus a walkthrough README.
Duration	1 hr

Dummy Dataset Files (ready to use)

Four starter files were generated alongside this plan so Phases 1–5 can start immediately:

- **employee_roster.csv** — 10 employees, 3 departments, reporting managers and cost centers (Phase 1, used again in Phase 5 for approval routing).
- **policy_rulebook.json** — 10 structured policy rules (Phase 1, consumed in Phase 3).
- **approval_matrix.json** — routing thresholds (Phase 1, consumed in Phase 5).
- **raw_receipts.json** — 12 dummy receipts with seeded edge cases (alcohol violation, duplicate pair, ambiguous scan, inflated outlier) for Phases 2–4.